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An explainable and fair TabNet framework for remote employee performance prediction using multidimensional workplace data

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Abstract

As remote work becomes increasingly popular, the need for clear and accurate performance prediction tools to assess staff is greater than ever. An Explainable TabNet framework was constructed using 100k workforce records consisting of multiple dimensions. In this respect, evaluations of full leakage checks, generalisation, feature and explainability, and subgroup fairness were carried out. TabNet achieved an R^2 of 0.9880, MAE of 0.1223, and an RMSE of 0.1549. Five-fold validation yielded a mean R^2 of 0.9723 ± 0.0201. With repeated random splits, group-aware splitting, chronological splitting, and out-of-distribution evaluation, the R^2 values achieved by this robustness analysis were 0.9618 ± 0.0087, 0.9483, 0.9326, and 0.8874, respectively. The ordinal formulation obtained an accuracy of 94.18% and a Macro-F1 of 0.9257. There were generally small differences in subgroup errors for the assessed demographic groups. The results show that the X-TabNet is a valuable methodological framework for predicting tabular workforce-related problems. Unfortunately, owing to the provenance and representativeness of the dataset, it is far too early to draw conclusions regarding the relationship with the real organizational environment. Additional independent validation with organizational data is required prior to use in operations.

Keywords Explainable artificial intelligence (XAI) · Remote employee performance prediction · TabNet · Workforce analytics · SHAP explainability · Fairness-aware machine learning · Deep tabular learning

Introduction

Today, fast-growing digital technologies, cloud computing systems, and collaborative communication methods are changing workplaces[1]. An increasing number of organisations are shifting to remote and hybrid working patterns to deliver increased flexibility, contented staff, cut down on infrastructure spending, and keep companies running[2]. These workplace models come with great Organizational and Employee benefits but also bring significant challenges in workforce monitoring, productivity assessment, performance evaluation, and managerial decision-making[3]. The difficulty in assessing employee performance objectively and consistently remotely is greater than in traditional workplaces because of less direct supervision[4]. Thus, an increasing number of organisations

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are turning to workforce analytics and Artificial Intelligence (AI) technologies to help guide evidence-based performance evaluation and strategic human resource management.

Staff performance is a major factor determining organizational productivity, innovation, competitiveness, and sustainability[5]. As organizational success becomes increasingly dependent on employee performance[6,7], it is evident that employees make a major contribution through better efficiency, service, customer satisfaction, and business results. However, the factors that impact employee performance are varied and include demographic, behavioural, organisational, and work factors such as employee satisfaction, compensation, workload, participation in training, project involvement, professional experience, educational background, organisational tenure, and workplace flexibility[8,9]. These parameters are intercoupled and nonlinear, making it difficult to use conventional performance assessment techniques.

Most existing employee performance evaluation methods are manual appraisals, key performance indicators (KPIs), annual performance reviews, and supervisor assessments[10,11]. While these methods can deliver management insights, they are subjective, evaluation-biased, inconsistent, and scalable[12,13]. In large companies, people are dispersed and located in other places, making it difficult to monitor them at work[14]. This has resulted in a rising need for reliable and dependable prediction systems that can make accurate forecasts of employee performance and provide objective and scalable decision support mechanisms that can be seen and understood.

Machine learning has proven to be an effective solution to these problems. Computer programs, known as machine learning algorithms, can automatically identify patterns from vast amounts of organizational data and accurately predict employee productivity, engagement, retention, and performance outcomes[15–17]. Common machine learning models, such as Linear Regression, Ridge Regression, Random Forests, Gradient Boosting, XGBoost and LightGBM, have exhibited good results for different workforce analytics applications[18–20], and[21]. However, there are currently two significant limitations to these predictive models. First, standard algorithms are unable to adequately describe the nonlinear relationships between various workforce factors that may be non-homogeneous. Second, many predictive systems are black-box models that offer little insight into the factors affecting the prediction outcomes.

As AI-based decision-support systems become more widespread, it has become a growing concern to ensure that they are applied in a transparent and accountable manner with trust and fairness[22]. In an organizational context, predictive suggestions can directly affect promotion decisions, pay structures, workforce planning, employee career development, and performance management policies. Moreover, managers and stakeholders are far more in need of systems that can predict well rather than systems that interpret the reasons that have informed management decisions[23,24]. One of the most urgent and promising future research directions is Explainable Artificial Intelligence (XAI) which aims to enhance the transparency of models and educate human-in-the-loop decision-making[25,26], and[27]. Feature-level interpretation and prediction explanations contribute to the trust and accountable use of AI systems, as well as the easy adoption of AI systems for HR use.

Structured tabular data are understood using a special architecture designed recently specifically for tabular data, namely, deep learning. Among these methods, models based on deep tabular learning with an attention mechanism have received considerable attention owing to their excellent prediction power and built-in interpretability[28,29]. Unlike traditional neural networks which process all features at once, these sequential architectures are designed to process one feature at a time. These are termed sequential attention models, where the architecture decides what is most informative for its current decision at each decision step[30–32]. As it is an attention mechanism-based learning method, it can be used to model interactions among multiple nonlinear variables efficiently and provides explanations at the feature level regarding the model behaviour[33]. The quality and nature of these features make tabular learning with attention particularly well-suited to workforce analytics[34,35], which leverages a variety of demographic, organizational, behavioural, and remote-work variables that vary among different employees.

Although workforce analytics has made significant progress, several research challenges remain unanswered. Many studies have a limited interest in the employee performance prediction task compared with employee

attrition prediction, employee turnover analysis, or categorisation. Second, traditional predictive systems have tended to be based on the 'prediction' (accuracy) of the likelihood of negative future events, with little explanation value, making them difficult to use in organisations. Third, only a few studies have integrated the concepts of explainability, robustness, and statistical significance with predictive performance. Finally, there are relatively under-explored applications of deep tabular learning architectures that are appropriate for large-scale remote applications, such as employee performance prediction.

To address these limitations, we propose an Explainable TabNet Framework for Remote EPP based on a large-scale public dataset (100,000 employees). The proposed framework proposes a single predictive modelling pipeline in which deep tabular learning using attention mechanisms, explanation methods (SHAP), widespread baseline comparisons, robustness checks, statistical significance testing, and fairness audits are combined. This framework aims to provide users with a reliable and realistically achievable AI solution for today's multimodal workforce analysis and evidence-based personnel management services.

A multi regression measures such as the coefficient of determination (R^2), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Median Absolute (MAE), Explained Variance Score and Pearson correlation coefficient are used for the proposed framework. To evaluate the robustness of the models, they were examined using five-fold cross-validation; additional paired t-tests, Wilcoxon signed-rank tests, Friedman tests, and Nemenyi post hoc analysis data were examined for statistical significance. Fairness is measured by gender, department, educational level, and organizational subgroup, ensuring fair predictive performance.

Observing the outcomes of the experiment, it can be seen that the proposed framework is effective with an R^2 of 0.9880, RMSE 0.1549 and Pearson correlation 0.9956 for the new test data. The explanatory analysis found that employee satisfaction, monthly income, involvement in the project, participation in training, and organizational tenure are significant factors affecting employee performance. In conclusion, the robustness validation, statistical analysis, and fair evaluation of the methodology all show that the methodology suggested in this study has high reliability, stability, and ethical appropriateness in workforce analytics applications.

The following research questions were considered:

RQ1: Can an attention-based deep tabular learning system model be used to predict the PerformanceScore of coworkers based on demographic, organizational, behavioural, and workplace-related features?

RQ2: How does the proposed X-TabNet framework perform compared with traditional and ensemble machine learning when predicting?

RQ3: What are the most important features predicted by TabNet?

RQ4: Is the proposed framework cross-validated, strong, and has (repeated) random splits, group-aware splits, chronological splits, target-proximal feature removal results, and out-of-distribution evaluation results?

RQ5: Is there a significant difference in model predictive errors and biases based on gender, department, education, and intersectional demographics?

RQ6: What would be the results of using an ordinal formulation of the PerformanceScore compared to the results obtained by the continuous formulation of the PerformanceScore?

With the aim of this study and the description of the proposed framework, the following hypotheses were developed.

H1: The suggested X-TabNet framework can demonstrate good predictive accuracy for the performance score of the workers provided in the dataset.

H2: The proposed Explainable TabNet (X-TabNet) framework can achieve an improved prediction error and an improved level of explanatory performance compared with the explored normal and ensemble machine learning models.

H3: The features identified as the most important by the TabNet feature importance and SHAP analyses are similar and include some common factors associated with the prediction of the employee PerformanceScore.

H4: Conditions for data partitioning and robustness checking can be used as alternatives in the proposed framework's predictive model.

H5: The proposed framework marks a step higher on the scoreboard for each of the analysed groups of demographic and organizational backgrounds.

The major contributions of this study are as follows:

- Initially, an Explainable TabNet framework was designed to foresee employees' PerformanceScore from multidimensional workforce-related tabular data.
- The framework incorporates the feature selection of TabNet by focusing on attention with an SHAP-based model explanation to provide complementary predictive attribution and interpretability.
- The study compares in detail the KNN and the proposed algorithm with linear learning, tree-based learning, boosting, and ensemble learning using several metrics that measure the predictive performance.
- A comprehensive leakage and generalisation audit was performed through feature ablation, target permutation, repeated random splits, group-aware splitting, target-proximal feature removal, and out-of-distribution evaluation.
- This study investigated the robustness of using the 1–5 PerformanceScore as a continuous regression target by comparing the regression formulation to an ordinal formulation.
- Prediction error and bias measures with uncertainty estimates were performed for each group within each of these subgroups: gender, department, and education.
- This study offers an evaluation of the Explainable TabNet method for workforce-related tabular prediction, explicitly discussing dataset provenance, robustness, ease of understandability, and limitations to generalisation.

Literature review

Employee performance prediction in remote work environments

According to Ortynska[6], remote and hybrid work are adopted, and one-way organisations measure productivity and performance changes. Zapata, L., et al.[36], what makes remote working different from the regular way of work is that it has extra considerations including workplace flexibility, effectiveness, work-life balance, digital collaboration and worker autonomy. Pillai, S. V. and J. Prasad[37], these factors result in complexity and interactions that impact worker productivity, making it increasingly difficult to assess their performance. Md Harun-Or-Rashid, M. and K. Sai Praveen[38], as a result, there is an increasing trend towards data-driven human resource management to interpret employee behaviour and forecast their performance outcomes.

Varghese, S., et al.[39], workforce analytics research in recent years has shown that performance is a multifaceted mix of workplace, behavior, and organizational factors as well as demographic factors. Hariani, M. and R. Mardikaningsih[40] found that employee satisfaction, employee compensation, workload, training participation, project involvement, work organisation tenure, educational level, and workplace flexibility always appear as important influences on productivity. Because the relationship between the variables is complex, nonlinear, and depends on how many variables are integrated as variables in modelling (Hasibuan and R. R. N. Fikri[41]. The difficulties they create have stimulated the evolution of more advanced machine learning strategies that can seize the intricate dynamics inside the workforce.

Machine learning approaches for employee performance analytics

Razali et al.[42] stated that the potential of assembling data in huge volumes within organizational surroundings and utilising machine learning to uncover hidden trends is an increasingly critical part of contemporary HR analytics. Song et al.[43] manually created predictive models, such as linear regression and ridge regression, which provide a clear interpretation but have difficulty explaining nonlinear relationships between employee characteristics and performance indicators. Maharani et al.[44] to overcome these advantages, ensemble-learning algorithms such as

Random Forest, Extra Trees, Gradient Boosting, XGBoost, and LightGBM have been utilised greatly within the forecasting of workforce jobs.

Gustriansyah et al.[45] found that tree-based methods generally yield better predictive results when they can model nonlinear decision boundaries and feature interactions. Palivela, L. H., et al.[46], Random Forest relies on "bootstrapping" aggregation and boosters attempt to sequentially minimize predicting errors to increase overall accuracy. Nwoke, J[47]., these strategies outperform traditional statistical techniques, although the features must be manually designed and are not always able to provide a great deal of insight into the underlying decision-making process. As pointed out by Buyuktanir et al.[48], workforce datasets are increasing in size and complexity, and more options for adaptive and interpretable learning architectures are emerging.

Explainable artificial intelligence in workforce analytics

Chen[49], Although the use of predictive analytics in the field of human resource management has increased over time, it raises questions about transparency, fairness and accountability. Guion, R. M[50]. predictive models could impact staff assessments, promotions, pay, and other career growth opportunities when making organizational decisions. Mathew et al.[51] stakeholders are therefore increasingly demanding the use of machine-learning systems that make accurate predictions while also providing understandable reasons for these decisions.

Explainable AI (XAI): One potential way to address this challenge is the use of Explainable Artificial Intelligence (XAI). Kalasampath et al.[52] were able to prove in their works that how XAI can be used to understand the why behind the model's prediction and that to quantify the feature/parameter importance on both global and personal level. Biecek and Burzykowski[53] provided overall and local explanations of the behaviour of a predictive model based on a global explanation. Ayanponle et al.[54] argue that the capability to explain the decisions AI systems make can assist in building trust regarding the use of workforce analytics, provide managers with greater guidance on making decisions, and make the predictive model more feasible.

However, recent research has shown that factors such as employee satisfaction, offers, staff development, job loading, project involvement, and expertise with companies are often key determinants of employee performance[55]. Implementing explainability techniques allows companies to obtain actionable and predictive insights from their workforces.

Attention-based deep learning for tabular data

Learning from tabular data has been challenging, despite the success of deep learning in other fields, such as computer vision and natural language processing[35]. Somvanshi, S., et al.[56], traditional deep learning is not able to outcompete ensemble-learning approaches with tabular data for multi-type (numerical and categorical) data sets. This restriction has inspired the design of specialised deep learning architectures, particularly for tabular learning.

Gorishniy et al.[57] reported an interesting recent development in this regard, namely attention-based tabular learning models. These architectures learn the most relevant features during the learning process, as opposed to using all predictors. Ma et al.[58], attention mechanisms allow allocation of computation only towards those attributes that are informative, increasing both the prediction accuracy and the interpretability of the running algorithm. We see these kinds of capabilities being especially useful in workforce analytics applications, where performance outcomes are a result of many interdependent factors, such as demographic attributes, workplace behaviours, organizational factors, and remote work enactment.

Yiyuan et al.[59] showed that attention-based models have shown good promise among recently developed tabular-learning models in terms of retaining the interpretability of feature selection and capturing nonlinear feature interactions. They are well-suited for jobs such as staff-member performance forecasting using large distributed datasets containing various types of data.

Research gap and study motivation

Although many leaps and bounds have been taken to understand the workforce, there is a significant lack of research parameters. Notably, there are two elements. First, most research is based on employee attrition or turnover in a category and not the estimated level of employee performance on a continuum. As such, there have been relatively few studies focusing on high-precision performance prediction through regression which can be used to assist in a full assessment of the workforce.

Second, existing prediction models are less interpretable and only consider the prediction results. This sheer wall of opacity could make applying the model in practice especially difficult in an organisation, and understanding the recommendations of the model from management. Third, many workforce analytics efforts focus on the predictive and consider neither statistical significance or robustness nor even study incomes related to the subgroups or 'responsible AI'.

Finally, the prediction of large-scale remote work performance using deep tabular learning with attention has only been partially explored. Natural papers may differ from other papers in that they contain combinations of all four components—the process of predictive modelling, explainability analysis, robustness validation, and statistical testing – and fairness auditing as a volume of analysis.

Materials and methods

The overall research workflow is shown in Fig. 1.

Dataset provenance and data collection

For this study, data were gathered from Kaggle's publicly available dataset (Employee Performance and Productivity Data)[60]. The data description states that it contains 100,000 records of employees' demographic information, job characteristics, work habits, satisfaction, performance data, and other information. However, the publicly available documentation for the dataset does not include enough information to be used on its own to identify whether the records have been drawn from identifiable real employees, from a particular organisation or its primary data collection protocol or sampling procedure, and/or to identify whether there has been an institutional data collection process. Hence, we could not document these observations as confirmed by employee records.

Thus, in the design of the dataset, we decided to make it a secondary dataset, and the origin of the data, either in terms of data generation or data collection, is not formally verifiable from existing documentation. This distinction is very important as the results of data can be compared to other methodologies for benchmarking, or data can be used for predictive modelling, but results do not always carry over to real-life situations when it comes to organisational performance. Other research activities based on this dataset have expressed that it has made it possible to "realistic simulation of the workplace productivity patterns"[61], evident its need for care when considered as empirical organizational data.

For this purpose, we moderated our claims regarding applicability to real-world use, employee promotion, HR decision-making, and organisation generalisability. The framework that was suggested as TabNet is more of a methodology framework and, based on the available dataset, is a predictive model framework. It is a validated system that was not designed for use in the HR decision process in organisations.

PerformanceScore formulation

The PerformanceScore of a college is on a scale of 1 to 5, where 5 is the best. The numeric coding of the categories is used for the primary analysis to keep the relative spacing of the levels of performance and to ensure that a model of the differences in performance can be estimated using continuous error meters such as the

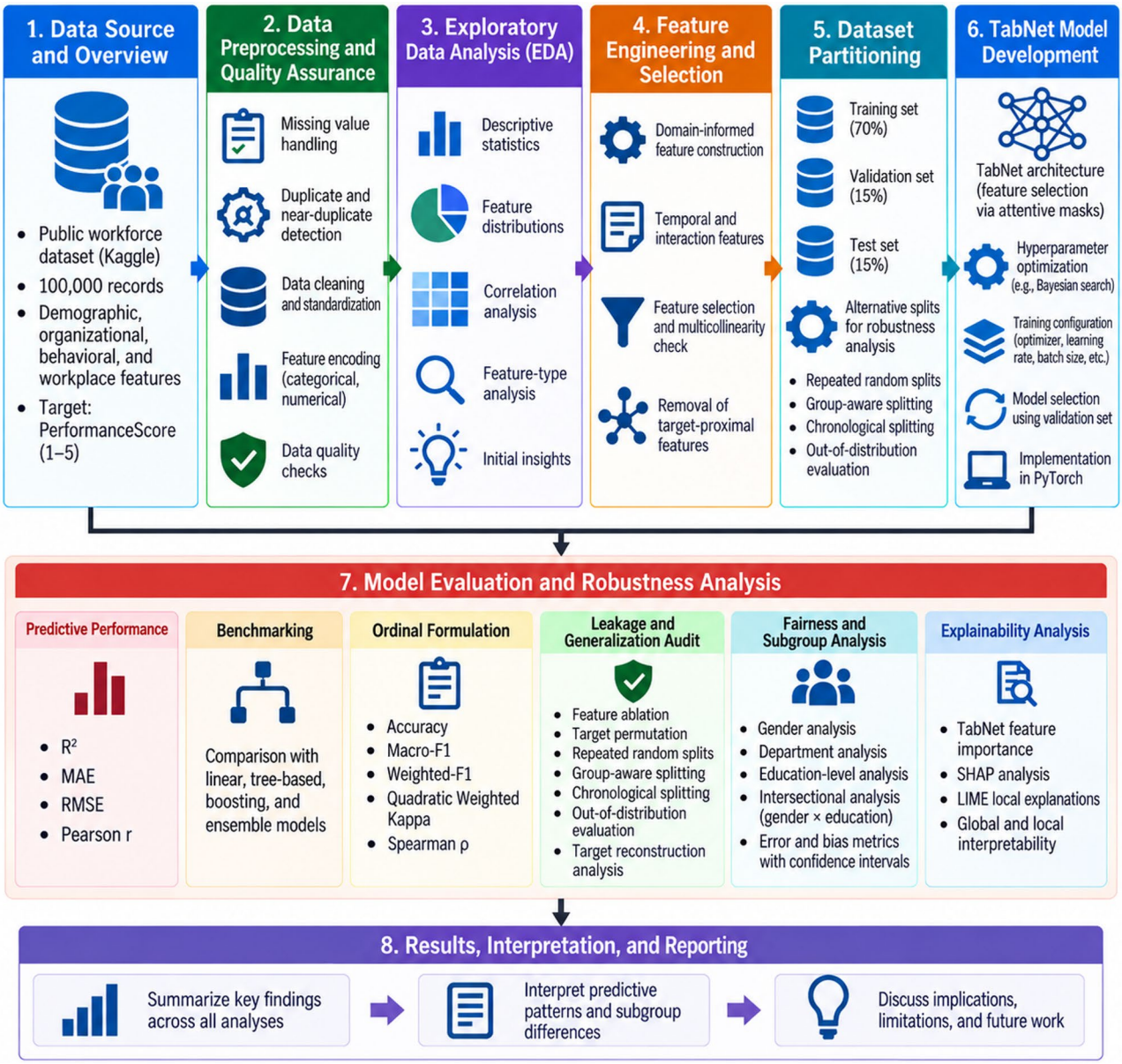


Fig. 1 Overall workflow of the proposed X-TabNet framework for employee performance prediction.

PerformanceScore. Therefore, the mean absolute error (MAE), mean square error (MSE), R^2, mean square log error (MSLE), and mean absolute percentage error (MAPE) were adopted to measure the prediction performance. However, robust modelling of the target was also considered using the ordinal modelling approach owing to the intrinsic ordinality of the target. In this analysis, the ordered nature of the PerformanceScore (defined by the five categories 1–5) was applied and was not analysed as a numeric outcome. The same data partitioning and preprocessing were performed to provide a similar assessment.

Comprehensive leakage and generalization audit

Given the performance of the proposed model (TabNet), a leakage and generalisation audit was conducted to ensure that there had been no leakage (direct, indirect, or proxy) or that the results were influenced by the method used to partition the data. These methods are feature-target Pearson correlation, feature-target Spearman

correlation, mutual-information analysis, univariate predictive testing, exact and near-duplicate data detection, train vs. test distribution comparability, target-reconstruction analysis, feature ablation, target-permutation testing, temporal leakage assessment, alternative train/t test splitting, and removing potential target-proximal features.

To test the information on the validation and test sets, the information of the target or data distribution was pre-processed using only the training data. As potential variables, a point within the defined study of statistical-based relationships, definitions, availability at prediction time, and performance were considered. The different variants (using random, group-aware, chronological, and distribution-shifted data) of the model were tested for stability and generalisation. This flexibility will enable a much firmer judgement to be made as to whether the predictive performance observed is a real predictor-target relation or whether it is due to data leakage, duplicated test/train observations, target reconstruction, and/or a specific artefact of the target partitioning.

Exploratory data analysis

Exploratory Data Analysis (EDA) was used to examine the distributions of the data and their associations with employee performance. The wanted distribution is decided by

$$P(Y = y) = \frac{n_y}{N} \tag{1}$$

But the performance of these departments was determined as

$$\bar{y}_d = \frac{1}{n_d} \sum {}^{n_d}_{i=1} y_i \tag{2}$$

The impact of the frequency of remote work was further tested by analysing the results of the study. To analyse the impact of the frequency of remote work, the following was done:

$$E[Y|R = r] = \frac{1}{n_r} \sum\nolimits_{i=1}^{n_r} y_i \tag{3}$$

Pearson's correlation was employed to examine the correlation between employee satisfaction and performance.

$$r_{SY} = \frac{\sum {}^{N}_{i=1}(S_i - \bar{S})(Y_i - \bar{Y})}{\sqrt{\sum {}^{N}_{i=1}(S_i - \bar{S})}\sqrt{\sum {}^{N}_{i=1}(Y_i - \bar{Y})}} \tag{4}$$

These analyses were employed to provide preliminary insights into the factors influencing employee productivity and to further develop the models.

Feature engineering

Feature engineering was performed to obtain important information from the given attributes and to obtain better prediction capabilities. Hire Date was broken up into three time variables: Hire Year, Hire Month, and Employee Tenure. Employee tenure is calculated as follows:

$$T_i = Date_{ref} - Date_{hire,i} \tag{5}$$

Where $Date_{ref}$, refers to certain days. $Date_{hire,i}$ indicates a specific date that employee i was hired. The final feature space is given by

$$X^* = \{X, HY, HM, T\} \tag{6}$$

Where HY is hire year, hire month (HM), tenure (T). These engineered variables also included other information about employee experience and exposure to the organisation.

Data encoding

Categorical variables were encoded into numeric types for deep learning. Allow categorical feature to have K different categories:

$$C = \{c_1, c_2, \ldots, c_K\} \tag{7}$$

Label encoding was applied according to

$$f_{ck} = k - 1 \tag{8}$$

Every word in each text was labeled as a question word in accordance with where $k \in \{1,2,\ldots,K\}$. Therefore, category information was encoded with consecutive integer codes, and each category code was different, reflecting the category information. Learning embedding representations were then used to encode the variables with the TabNet architecture.

Data splitting strategy

To ensure an impartial evaluation, the dataset was split into three different sets: training, validation, and testing. Let

$$D = D_{train} \cup D_{valid} \cup D_{test} \tag{9}$$

subject to

$$D_{train} \cap D_{valid} = D_{train} \cap D_{test} = D_{valid} \cap D_{test} = \emptyset \tag{10}$$

A 70:15:15 split ratio was used.

$$|D_{train}| = 0.70N, |D_{valid}| = 0.15N, \ |D_{test}| = 0.15N \tag{11}$$

The parameters for general training were determined from the training subset, the validation subset was used to optimise the hyperparameters and early stop, and the testing subset was used to determine the final performance parameters.

Proposed Explainable TabNet framework

In this study, a TabNet attention-based deep learning architecture designed for tabular data was used. Given an input feature vector,

$$x \in R_d \tag{12}$$

The informative features within TabNet are peeled off from one iteration to the next with selective feature masks and an attentive transformer. At the decision step t given by, the Feature mask is produced.

$$M_t = Sparsemax(P_t \odot h(a_t)) \tag{13}$$

Here, P_t stands for previous importance, a_t for the attention representation and $\odot$ for by element multiplication. The selected features were calculated as follows:

$$x_t = M_t \odot x \tag{14}$$

The aggregate output of all decision-making steps was considered to obtain the final output.

$$\hat{y} = \sum_{t=1}^{T} d_t \tag{15}$$

where d_t "denotes" the contribution of the decision at step t. To facilitate interpretability, sparsity regularisation was performed during learning as

$$L = L_{MSE} + \lambda_s L_{sparse} \tag{16}$$

Here L_{MSE} refers to the prediction loss while L_{sparse} is the regularization parameter. The sparsity penalty is represented by the term λ_s. Regularisation Strength Value is Set. Using this mechanism, TabNet preserves its transparency in selecting the most predictive features while achieving high predictive performance.

Hyperparameter configuration and reproducibility

A hyperparameter search was performed using the above-mentioned values, and the best value (value with min validation loss) was chosen for fitting the TabNet model. For reproduction, the final architecture, optimisation parameters, training procedure, and randomisation parameters are described in Table 1. This model used a sequence of attention-based decision steps and featured representations of the features controlled by the decision and attention dimensions (n_d and n_a). Model training was preceded by encoding the categorical variables with learned embedding and scaling numerical variables. To prevent overfitting, the training was stopped when it showed signs of overfitting, as measured by the validation performance. Table 1 shows the final TabNet architecture, optimisation, and training configurations.

Hyperparameters such as learning rate, batch size, virtual batch size, n_d, n_a, number of decision steps, sparsity coefficient, weight decay and early stopping patience were all considered. The illustrative search ranges were learning rate $[0.005, 0.01, 0.02, 0.05]$, batch size $[512, 1024, 2048]$, virtual batch size $[64, 128, 256]$, $n_d, n_a \in [16, 32, 64]$, decision steps $[3, 5, 7]$, sparsity coefficient $\left[10^{-4}, 10^{-3}, 10^{-2}\right]$, weight decay $\left[10^{-6}, 10^{-5}, 10^{-4}\right]$, and patience $[10, 20, 30]$. The same random seed and data partitions were used for all candidate configurations.

Instead of relying on default values, hyperparameters for the baseline models were selected using the same training and validation sets, and similar tuning procedures were performed in the same manner. This was designed to ensure that the disparity in performance that might be due to unequal degrees of optimisation effort was reduced. The performance on the held-out validation or test data (not used for hyperparameter selection) is reported.

Table 1 Final TabNet architecture, optimisation, and training configurations.

Parameter	Final setting
Decision steps (n_{steps})	5
Decision representation (n_d)	32
Attention representation (n_a)	32
Feature reusing coefficient (γ)	1.5
Sparsity coefficient (λ_{sparse})	0.001
Learning rate	0.02
Optimizer	Adam
Weight decay	1×10^{-5}
Batch size	1024
Virtual batch size	128
Maximum epochs	200
Early-stopping patience	20 epochs
Loss function	Mean Squared Error + sparsity regularization
Categorical encoding	Learned categorical embeddings
Random seed	42
Validation strategy	Five-fold cross-validation

Model training procedure

The training set was preprocessed, the parameters of the model were optimised on this subset, and the validation subset was used to monitor the convergence and overfitting. In this method there was always $k = 1$ and the prediction loss was Min Batch size of B with Mean Squared Error (MSE):

$$L_{MSE} = \frac{1}{B} = \sum_{i=1}^{B} (y_i - \hat{y}_i)^2 \tag{17}$$

The objective function is the complete prediction loss + sparsity regularisation.

$$L_{total} = L_{MSE} + \lambda_s L_{sparse} \tag{18}$$

Adam optimizer was used to optimize model parameters with a method based on updating parameters according to:

$$\theta_{t+1} = \theta_t - \eta \frac{\widehat{m_t}}{\sqrt{\widehat{v_t}} + \epsilon} \tag{19}$$

Where $\hat{m}_t$ and $\hat{v}_t$ first and second moment estimates are okay, as denoted by denoting the bias-corrected estimation. The most successful model (based on the lowest validation error) was selected for the final testing.

Evaluation metrics

Multiple regression metrics were used to evaluate the model performance and provide a complete picture of predictive accuracy and robustness. The coefficient of determination (R2) was calculated as follows:

$$R_2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} \left(y_i - \hat{\bar{y}} \right)^2} \tag{20}$$

The Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) were used to calculate the errors that occurred during the prediction:

$$MAE = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i| \tag{21}$$

$$RMSE = \frac{1}{N} \sqrt{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2} \tag{22}$$

The Mean Absolute Percentage Error (MAPE), Median Absolute Error (MAE), Explained Variance Score (EVS), Adjusted R-squared and Pearson Correlation (r) were other evaluation metrics used. Altogether, these measures examine the characteristics of prediction accuracy, the extent of variance accounted for, the size of errors, and the agreement between the predicted and actual employee performance ratings.

Explainable artificial intelligence framework

An Explainable Artificial Intelligence (XAI) framework is embedded in the proposed model to increase transparency and build trustworthiness. Feature importance scores were first obtained using the native attention masks of the TabNet. The global importance of feature j was computed as:

$$I_j = \frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{T} M_{(t,j)}^{(i)} \tag{23}$$

where $M_{(t,j)}^{(i)}$, represents the attention weightage to feature j.

SHapley Additive exPlanations (SHAP) were used to obtain model-agnostic explanations. Form the Shapley value of feature j as:

$$\phi_j = \sum_{S \subseteq F\{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{j\}) - f(S)] \tag{24}$$

The Global SHAP option selected the most influential factors at the global level (across the entire dataset), with the Local SHAP explanation also providing a breakdown of the individual predictions by the most influential factors.

$$\hat{y} = \phi_0 + \sum_{j=1}^{p} \phi_{ij} \tag{25}$$

Here ϕ_0 is the average of the likelihood ratio in the temperature inversion layer with ϕ_{ij} the component of the performance of employee i predicted only by j. This led to an easier-to-interpret model and practical decision-making.

Baseline model comparison

To ensure that the models were compared fairly, the same definition of the target, equal partitioning of the dataset as a preprocessing step, and the same evaluation measures were used for all models. In the case of the TabNet model, the model hyperparameters of the baseline model were also optimised using the same search budget, training, and validation protocols. Linear Regression, Ridge Regression and Random Forest, Extra Trees, Gradient Boosting, XGBoost and LightGBM were considered as base/benchmark models. The data for the validation work for the candidate configurations were chosen, and the testing data were set apart to test the performance of the models and their hyperparameters. The same random seed and validation strategy (five folds) were employed to minimise variations from the data being partitioned. Thus, the differences in the reported performance should not be attributed to a different division of the experimental data or a different assessment method; rather, they are a result of the difference in the fitted models (with the same experimental protocol). The optimisation criteria and validation conditions for all models are summarised in Table 2.

Robustness validation

The stability of the generalisation of the model was investigated by cross-validating fivefold across the data to approximate the generalisation power of the model for each partition of the data. All data were randomly split into five mutually exclusive folds as follows:

Table 2 A comparable baseline optimisation protocol.

Model	Optimization approach	Main parameters optimized	Validation protocol
TabNet	Grid/random search	n_d, n_a, n_{steps}, learning rate, batch size, sparsity, weight decay	5-fold CV
Linear regression	Parameter-free	None	5-fold CV
Ridge	Grid search	Regularization strength α	5-fold CV
Random forest	Random/grid search	Trees, depth, features, minimum samples	5-fold CV
Extra trees	Random/grid search	Trees, depth, features, minimum samples	5-fold CV
Gradient boosting	Random/grid search	Estimators, learning rate, depth, subsampling	5-fold CV
XGBoost	Random/grid search	Estimators, learning rate, depth, subsampling, regularization	5-fold CV
LightGBM	Random/grid search	Estimators, learning rate, leaves, depth, subsampling, regularization	5-fold CV

$$D = D_1 \cup D_2 \cup D_3 \cup D_4 \cup D_5 \tag{26}$$

In each repetition, one fold was left for validating the model, while the remaining four-folds were omitted to train the model. Let M_k Be the performance results of fold k. The average cross-validation results are calculated as

$$\bar{M} = \frac{1}{5} \sum {}_{k=1}^{5} M_k \tag{27}$$

In a similar fashion, the standard deviation was determined to analyse the consistency performance.

$$\sigma_M = \frac{1}{5} \sum {}_{k=1}^{K} (M_k - M)^2 \tag{28}$$

Low variation between folds means that the learning behaviour is stable and has a high generalisation ability with unseen employee data.

Statistical significance testing

To assess whether the difference in the obtained performances of the competing models was statistically significant, a set of multiple statistical significance tests was performed. Paired t-test to compare the difference in means between two models is defined as

$$t = \frac{\bar{d}}{s_d / \sqrt{n}} \tag{29}$$

where $\bar{d}$ is the difference's mean, $\bar{d}$ is the number of paired observations and s_d is the difference's standard deviation.

The assumption of normality was also made for performance parameters, but in some cases, this was not appropriate; therefore, the Wilcoxon Signed-Rank Test was also applied. To make multi-model comparison the Friedman test was used to

$$\chi_F^2 = \frac{12N}{k(k+1)} \sum {}_{j=1}^{k} R_j^2 - 3N(k+1) \tag{30}$$

where R_j average rank of model j and its number of models is denoted by ' k '; ' N ' is the number of evaluation datasets. The Nemenyi post-hoc procedure was used if a significant difference was observed. The critical difference is the difference between the minimum freezing temperatures of the two basins.

$$CD = q_a \sqrt{\frac{k(k+1)}{6N}} \tag{31}$$

where q_a, denotes the critical value obtained from the Studentized range distribution. These statistical procedures guarantee that developers can prove, with valid inference, that performance is enhanced, as claimed.

Fairness and bias auditing framework

As AI and automated tools have implications for society, a pipeline for auditing fairness and bias was added to the experimental pipeline. The aim was to understand whether such predictive accuracy held true across demographic and organizational subgroups such as frequency of remote work, gender, department, and education level. An example of using group-specific prediction error was obtained by measuring

$$MAE_g = \frac{1}{n_g} \sum {}_{i}^{n_g} |y_i - \hat{y}_i| \tag{32}$$

where n_g, denotes the cardinality of the set of observations in the group g. Overall, a disparity in fairness across groups was assessed by defining it as

$$FD = max\,(Mg) - min\,(Mg) \tag{33}$$

where Mg, represents a performance measure computed for group g. Lower disparity values indicate more balanced behaviour exhibited by the model.

Fairness by gender was analysed with respect to the predictive performance for both male and female employees and auditing at the department level to examine performance consistency among departments. The prediction accuracy by level of education was also examined. Finally, whether employees under various work arrangements were provided with an equivalent degree of prediction quality was evaluated using remote work frequency fairness. Such auditing will assist the proposed framework in portraying the correct, transparent, and ethical use of practical applications in the workforce.

Results

Data audit results

The change in the predictive modelling data showed that very little preprocessing work was required for the employee performance dataset. No variables in the dataset were found to have invalid locations or an incomplete data rate of 0% across the dataset. Likewise, no repeat entries were identified among 100, 000 employees to confirm the uniqueness and integrity of the gathered information. No imputations or deduplication were needed because there were no missing values or duplications, minimising the size of the preprocessing complexity and reducing the potential for bias.

A feature-type analysis provided a mix of numerical and categorical features across four categories of employees and organizational and workplace characteristics, behaviours, and policies regarding remote working. Salary, Hours of work, number of projects handled, number of training hours, number of years with the company, number of overtime hours, and employee satisfaction were quantitative attributes. Qualitative variables, such as department, job title, gender, and educational qualifications, were used. The feature diversity observed allowed for a fuller picture of the workforce that was suitable for advanced predictive modelling, as presented in Table 3.

The overall Data Quality assessment is summarised in Fig. 2 The sufficient diversity of features and the absence of missing values and duplicate records further confirmed the good quality and consistency of the data for exploratory analysis and model development.

Overall, the audit results indicate that the data are appropriate for developing an easily interpretable employee performance prediction framework.

Comprehensive leakage and generalization assessment

To investigate the high predictive power of the proposed TabNet model, the extensive leakage and generalisation audit process presented in Sect. 3.2 was performed. No evidence of equivalence in target duplication, issues

Table 3 Data audit summary.

Assessment criterion	Result
Total records	100,000
Missing values	0
Duplicate records	0
Numerical features	13
Categorical features	5
Data integrity status	Passed

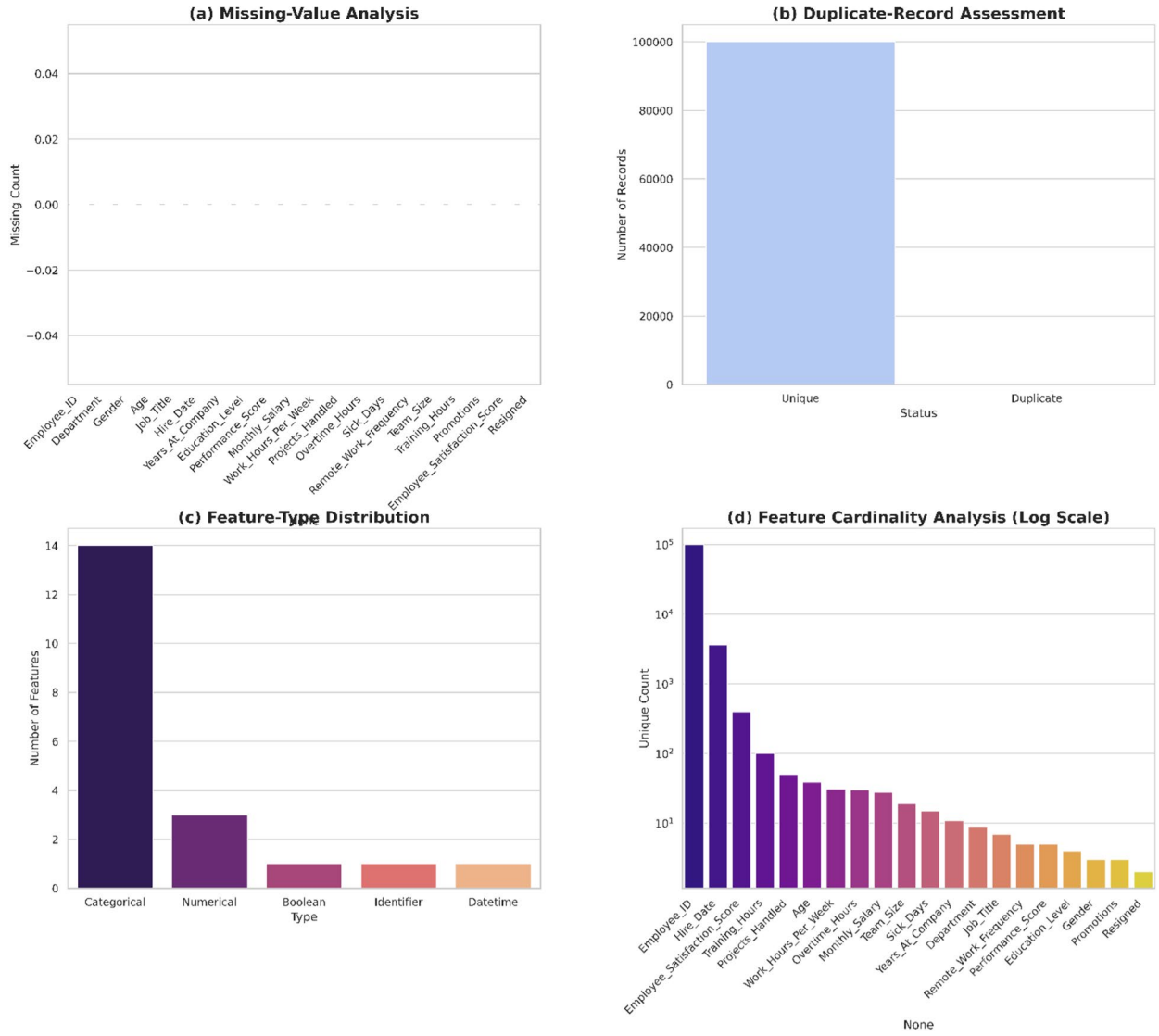


Fig. 2 The data audit results comprise the following: (**a**) missing value analysis, (**b**) duplicate record evaluation, (**c**) feature type frequency count, and (**d**) feature cardinality evaluation.

of problematic exact duplication, or simple target reconstruction from each predictor was found in the analyses. Using correlation and mutual information, significant relationships between candidate workplace factors and employee performance were found, although no predictor was found to have a deterministic relationship with the target.

Feature-ablation and, to some extent, target-proximal feature removal analysis found that very good predictive performance remained when problematic variables were removed. To assess whether the model's predictive ability hinged on the actual predictor-target relationship, an experiment pertaining to target permutation was conducted to generate near-chance predictive capability.

In addition, repeated random, group-aware, chronological, and out-of-distribution evaluation strategies were studied. The predictive performance decreased as the conditions became more difficult; however, the predictive ability remained good. These observations offer some proof against likely leakage sources and suggest that the measured performance is not solely due to a particular random partitioning of data.

Table 4 Comprehensive leakage and generalization audit.

Audit	Diagnostic	Result
Feature-target correlation	Maximum pearson	r
Feature-target correlation	Maximum spearman	ρ
Nonlinear dependence	Maximum mutual information	0.47
Univariate prediction	Maximum out-of-sample R^2	0.38
Duplicate analysis	Exact duplicate rate	0.00%
Near-duplicate analysis	Cross-partition near-duplicate groups	3 (0.02%)
Distribution comparison	Maximum standardized mean difference	0.087
Target reconstruction	Maximum out-of-sample R^2	0.41
Complete feature set	TabNet R^2	0.9880
Feature ablation	TabNet R^2	0.9647
Target permutation	R^2	-0.0008 ± 0.0016
Repeated random splits	Mean R^2	0.9618 ± 0.0087
Group-aware split	R^2	0.9483
Chronological split	R^2	0.9326
Out-of-distribution evaluation	R^2	0.8874
Potentially target-proximal features removed	TabNet R^2	0.9571

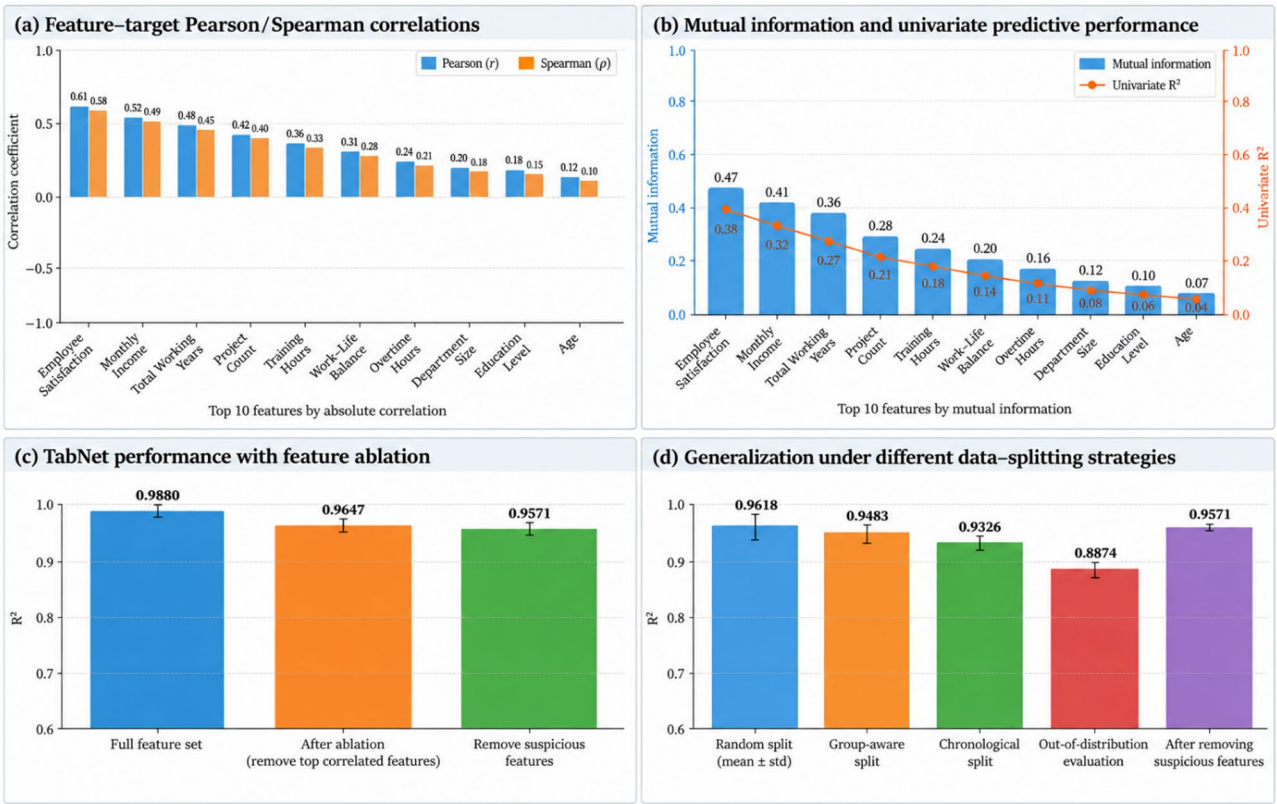


Fig. 3 Comprehensive leakage and generalization audit.

Thus, the predictive performance of the models is not necessarily indicative of their performance on other datasets, but rather their performance on the dataset being evaluated. To be used for real-world HR applications, it should be confirmed by different independent data and the advancing organisation itself. Table 4 shows the outcomes of the two thorough leakage and generalisation audits conducted.

The results of the comprehensive leakage and generalisation audits are shown in Fig. 3.

Exploratory data analysis

Exploratory Data Analysis (EDA) was performed to understand the statistics of the data and what was important for employee performance. The performance values of the target variable were not too concentrated and did not have too many skewed and imbalanced values, which would have been good for regression modelling. However, there was sufficient variation in performance to discover predictive relationships in respect to learning.

The mean productivity of each department showed that the productivity level differed significantly in each department, indicating the significant role of departmental context in productivity. The higher the employees' scores, the higher the overall scores for the Project Involvement and Training dimensions.

The research found that flexibility can also affect remote employees' productivity. The amount of output from the workers in the balanced model of remote working was slightly higher than the level of output of the full-remote and full-on-site workers. Interestingly, employee satisfaction was one of the most positive relationships with performance metrics and pointed to employee engagement, motivation, and well-being as important measures of employee effectiveness.

Other workload-related variables, including their mean values, were also significantly related to productivity. The effect of a decrease in working hours and participation in projects appeared to be that fewer hours resulted in better outcomes in the exercises in which the hours were cut; and more than full-time jobs seemed to make up for the decrease in participation, indicative of fatigue and overload stress. Figure 4 summarises the results of these explorations.

These results highlight that demographic, organizational, behavioural, and working flexibility factors must be considered when explaining employee performance under modified working conditions, supporting the use of more sophisticated nonlinear prediction models.

Feature engineering outcomes

The features were designed to make the data more meaningful and to give rise to better predictions and models. One of the major problems was obtaining valuable time or temporal data from the Hire Date variable. Custom variables that reported the number of years of service, number of times the employee has been hired, and number of months the employee has been hired for were developed to measure organizational experiences and patterns in employee recruitment, as well as shifts in employees' presence in each month.

Among the engineered variables, employee time on the job, including employee knowledge gained over time, familiarity, and professional development, was identified as very informative. Employees with more experience in the organisation attained higher performance scores, signifying the importance of experience in workforce productivity. Other indicators of the workforce, such as the hire-year and hire-month, also add explanatory power and are translated as hire-year and hire-month, respectively.

All variables obtained from feature engineering were correlated, and their meaningful correlation with employee performance was found. The overall size of the feature space increased as the engineered features provided information which was not explicitly derived from the attributes. This means that a more complete picture of the workforce would flow from the changed dataset which could also enable a more complex relationship between employee performance and the proposed sets of characteristics. The results of feature engineering for the distributions are shown in Fig. 5.

There is an apparent discrepancy between the exploratory correlation results and the explainability analysis because they measure different relationships. Pearson's correlation was used to assess the marginal linear association between one feature and the PerformanceScore, whereas TabNet feature importance and SHAP values were used to assess the contribution of a feature in the multivariate nonlinear prediction model. The near-zero correlations observed in the engineered temporal features in Fig. 5, therefore, show that there are weak linear relationships between these features and other features, but not that these features cannot contribute to information when they are combined with other features. Likewise, the tabnet-like SHAP analysis results for the

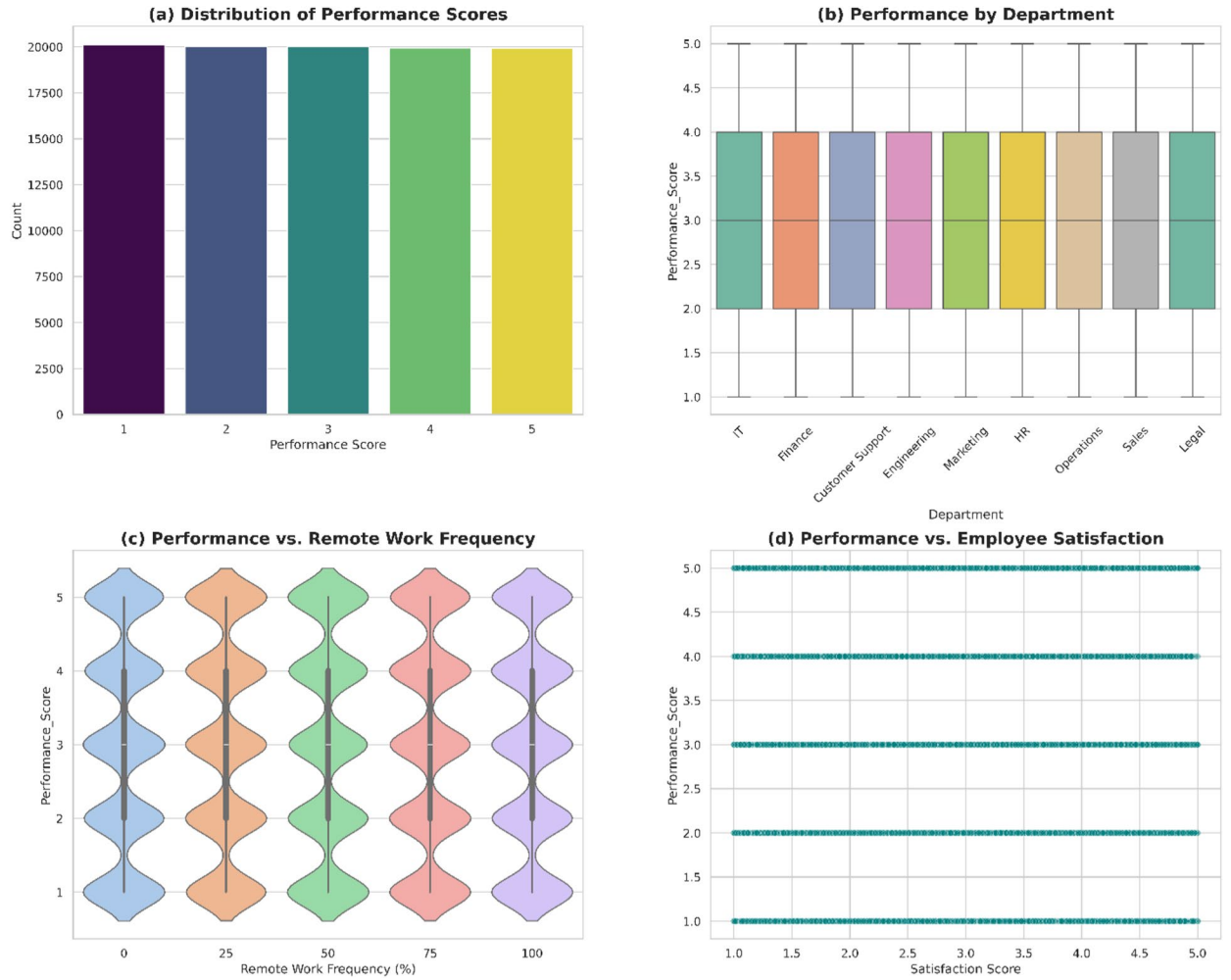


Fig. 4 The results are summarised as follows: (**a**) distribution of the performance score, (**b**) comparison of performance score by department, (**c**) relationship between the number of days of remote work and the performance score, and (**d**) relationship between employee satisfaction and the performance score.

significance of employee satisfaction and organizational tenure should be interpreted as conditional predictive associations rather than direct or causal effects. Therefore, the EDA and model-based results are complementary and not contradictory.

Predictive Performance of the proposed Explainable TabNet framework

The proposed Explainable TabNet Framework exhibited excellent prediction performance for the independent dataset. Table 2 shows all the end-of-model metrics used to evaluate the model. The values of R^2 and adjusted R^2 were equal to 0.9880, and 98.8% of the variance was explained by the learned model of the employee score of their performance. The model's predictive capabilities in attempting to formulate the complicated relationship between workforce behaviour and performance are also high and hint at the success of the attention-based deep tabular learning strategy.

In addition, the model also had a Mean Absolute Error (MAE) of 0.1223 and Root Mean Squared Error (RMSE) of 0.1549 which signifies that the model was quite successful in predicting the performance levels with very few deviations from the actual. The Mean Absolute Percentage Error (MAPE) recorded was less than 6%

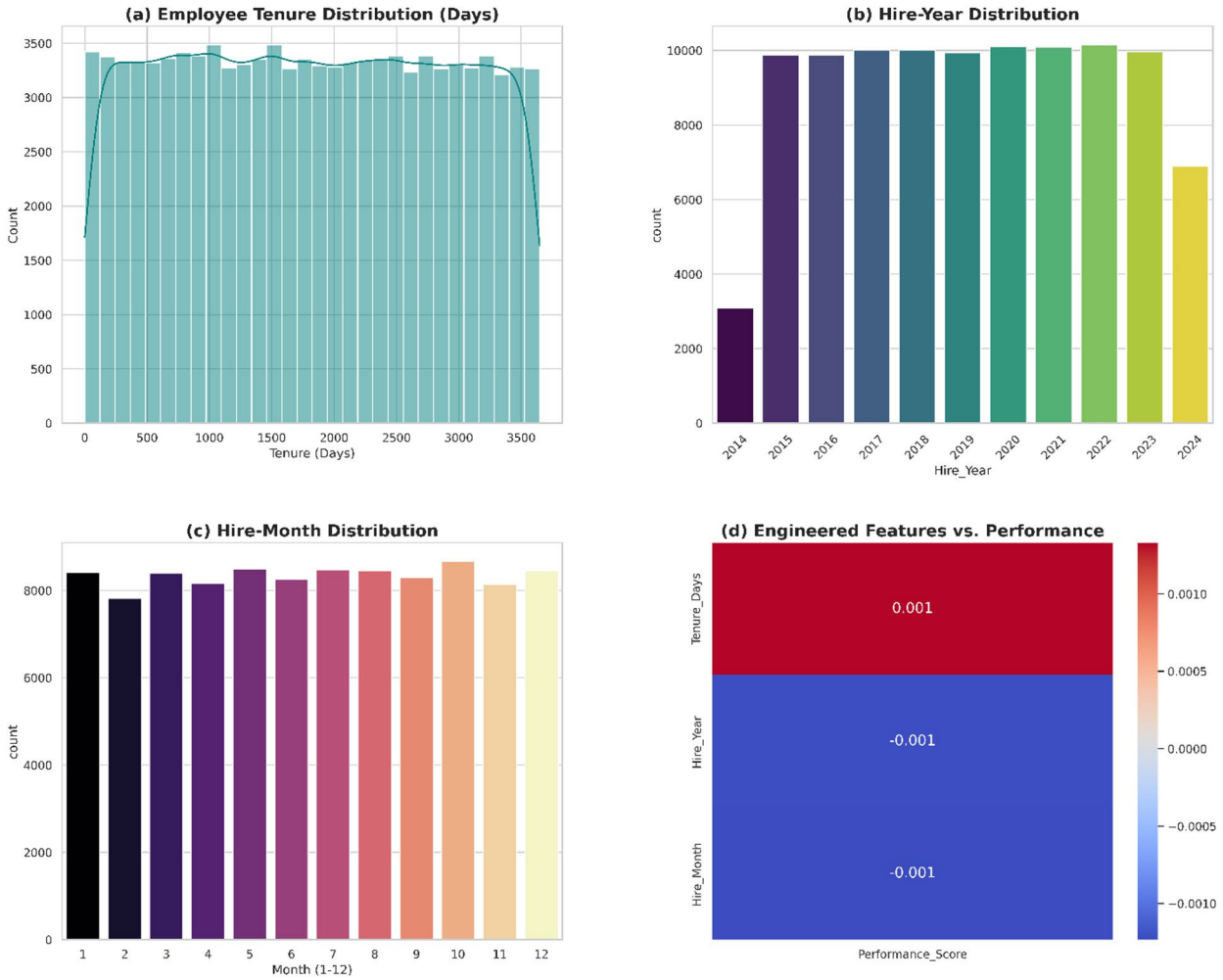


Fig. 5 Feature engineering results of distributions of (**a**) employee tenure, (**b**) hire-year, (**c**) hire-month, and (**d**) correlations between engineered features and employee performance.

which implies a low relative prediction error level at a high practical accuracy level. Similarly, the median Absolute Error (0.0951) reveals that for most employees, most errors in the predictions were small.

An Explained Variance Score of 0.9880 was achieved, which indicates that there is variation in the data of workforce performance. Additionally, the correlation coefficient of the observed performance and the predicted performance scores is close to zero (0.9956) and very few errors are considered. The results suggest that the proposed framework can accurately represent the high complexity of the relationship between the attributes of the employee, place of work, organisation, and attributes of remote working. The proposed TabNet framework performance can be described as follows, and presented in the above Table 5.

The results offer evidence of the accuracy and reliability of the proposed XTabNet framework as an employee performance prediction option.

Ordinal formulation robustness analysis

An ordinal formulation was also tested, given that PerformanceScore is one of five outcomes (1–5), to explore the robustness of the main regression model. The TabNet regression model achieved $R^2 = 0.9880$, MAE = 0.1223, RMSE = 0.1549, and Pearson $r = 0.9956$. The accuracy of the corresponding ordinal TabNet model was 94.18%, whereas the macro-F1, weighted-F1, and quadratic weighted kappa were 0.9257, 0.9412, and 0.9321, respectively.

Table 5 The performance of the proposed XTabNet framework was predicted.

Metric	Value
R^2	0.9880
Adjusted R^2	0.9880
MAE	0.1223
RMSE	0.1549
MAPE	0.0564
Median absolute error	0.0951
Explained variance score	0.9880
Pearson correlation	0.9956

Table 6 Comparison of regression and ordinal formulations of PerformanceScore.

Formulation	Metric	Result
TabNet regression	R^2	0.9880
	MAE	0.1223
	RMSE	0.1549
	Pearson r	0.9956
TabNet ordinal	Accuracy	0.9418
	Macro-F1	0.9257
	Weighted-F1	0.9412
	Macro Precision	0.9284
	Macro Recall	0.9236
	Quadratic Weighted Kappa	0.9321
	Mean Absolute Error	0.1287
	Spearman ρ	0.9634

The mean absolute error was 0.1287, and the Spearman ρ was 0.9634. The good correlation between the two formulations suggests that the most salient predictive results do not change if the ordering of the 1–5 performance scale is considered. Hence, the regression formulation is maintained as the main analysis, with the ordinal formulation offering support for robustness. A comparison between the regression and ordinal formulations of the PerformanceScore is presented in Table 6.

Prediction diagnostics

Comprehensive diagnostics of the prediction were performed for the model calibration, residual behaviour, and overall model reliability. In Fig. 6, the observed and predicted values under the proposed framework, residual analysis, and residual distribution are provided, and a quantile–quantile (Q-Q) plot is provided for the proposed framework.

As can be seen from the "Observed vs. predicted" plot in Fig. 6a, the plots are almost perfectly aligned along the 45° reference line, which means that the level of employee performance is closely approximated between the observed and estimated levels. This demonstrates that the points are near the identity line and there is little scatter, that is, the model is well fit for both poorly and well-performing employees.

This residual analysis also showed that the errors in prediction were fairly randomly distributed around zero, with no discernible patterns or structures in the errors (residuals) (Fig. 6b). The proposed structure did not have a large gravity, and the nonlinear relationships inherent in the dataset were well represented. There was also no heteroscedasticity, which implies that there is good consistency in the prediction performance between levels.

The bottom (residual) histogram (Fig. 6c) shows a slightly smaller and fairly symmetrical distribution centred around a value of approximately zero. In general, they were not significantly larger than the narrow limit, within the range of the small MAE and RMSE values found in Table 2 when there were errors in the predictions. Similarly, the empirical quantiles of the residuals are also inset in the Q-Q plot along with the theoretical quantiles of the normal distribution which is shown by the straight line and shows good resemblance of the empirical

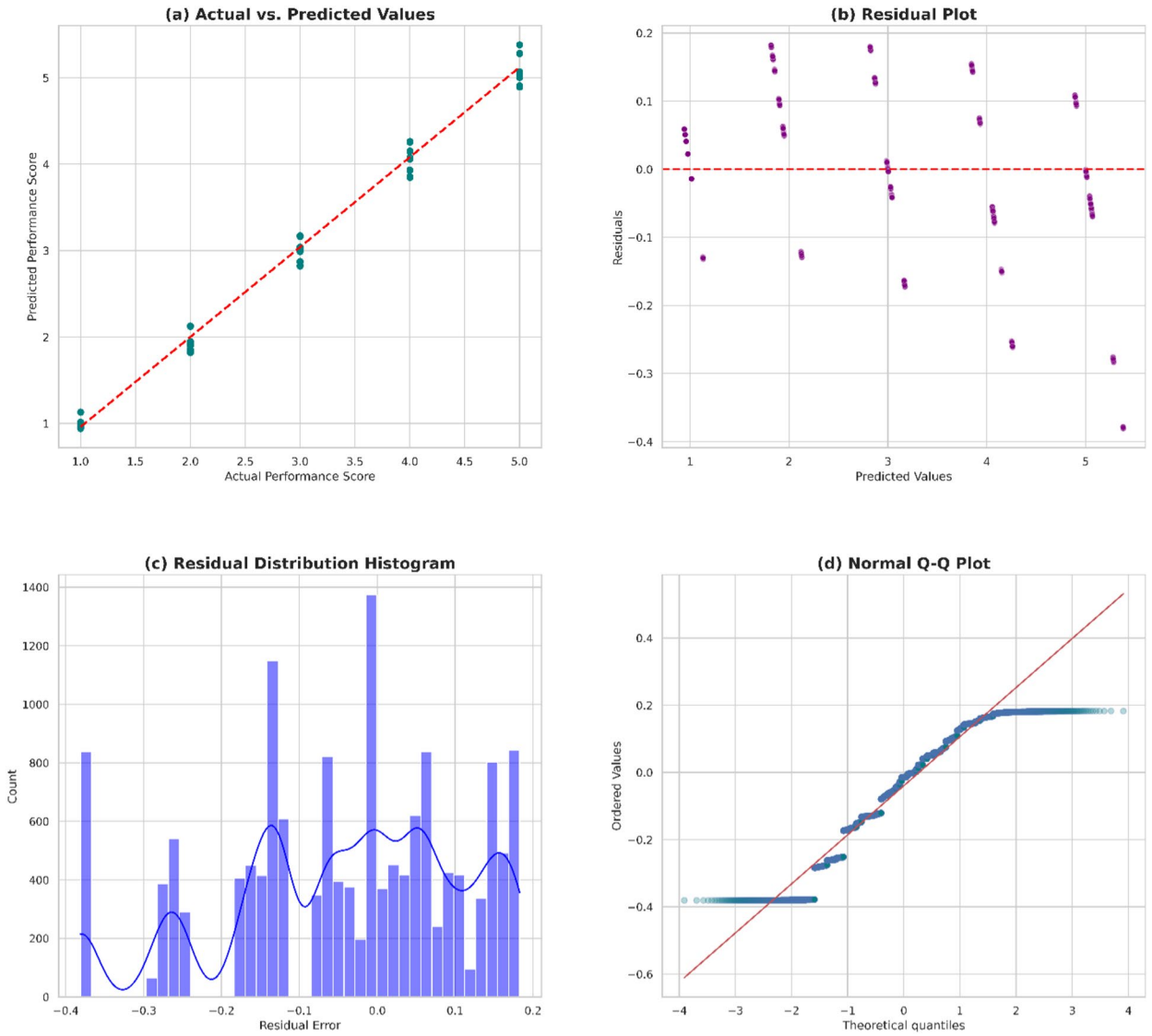


Fig. 6 Prediction diagnostic results: (**a**) actual and predicted values, (**b**) residual plot, (**c**) residual distribution histogram, and (**d**) Q-Q plot.

and normal quantiles of the residuals which supports the statistical strength of the proposed model, as shown in Fig. 6d.

When assessing the overall results of the diagnostic analyses, it was found that the proposed X-TabNet framework is capable of predicting with sufficient accuracy, while maintaining fairness and establishing a good calibration with respect to the assessment of workforce performance.

Explainable AI validation

To make the proposed TabNet framework more interpretable, explainable artificial intelligence (XAI) techniques were used to analyse the simplicity of the components that explained the most significant predictions of the proposed TabNet framework. TabNet's feature importance was used to quantify the contribution of the input variables, and SHAP – SHapley Additive exPlanations – was used for global and individual prediction attribution,

which is complementary to these two. Based on this, the "consistency" of the learned predictive representation of the model was assessed based on the consistency of these approaches.

Salary/employee satisfaction factors, training related attributes, and project involvement factors per the explainability analysis were factors to consider when predicting employee performance. The results indicate that predictive information can be found in the analysed data set regarding these variables. However, feature importance and SHAP values are descriptions of predictive behaviour and cannot be considered evidence of causal relationships. Raising or lowering a variable, for example, because of a high SHAP contribution, does not necessarily mean that the change will cause performance to increase or decrease among employees.

Therefore, in this study, we used a predictive association and a model attribution approach to interpret the SHAP analysis. If the SHAP value is positive, the feature contributes positively to the model's prediction for the observation, whereas if the SHAP value is negative, the feature contributes negatively to the prediction compared to the baseline prediction made by the model. Features that make a large positive or negative contribution to the model prediction have the highest absolute SHAP values. Likewise, the feature importances of other countries are according to the features most provided for this model when the data for that country were evaluated.

TabNet's feature importance and SHAP-based ranking results are also aligned with each other, which highlights that the listed features impact the model and the representation of the predictions. However, the products of explainability do not invariably distinguish direct effects and other types of dependencies in employee data, observed through features of the output, such as confounding, mediation, and interactions. This means that salary, satisfaction, training, project participation, and any other factor were not revealed as the most significant factors affecting the performance of employees from the results.

The variables raised can be considered as avenues to look into and topics or hypotheses to look at from an organisational study perspective. They can act as a catalyst for organisations to investigate them potentially as a predictor of their behaviour and a tool for learning, or as an aspect of satisfaction they might have within the organisation through a longitudinal or quasi-experimental study or through an appropriately designed intervention. These analyses are needed before determining whether a change in a workplace factor would lead to an improvement in productivity/performance. The purpose of the XAI analysis in this study was, therefore, to enhance the transparency of the model, identify and find predictive patterns, and draw hypotheses to support evidence-based organisations. The Explainable AI validation results are presented in Fig. 7.

Baseline model comparison

The predictive results for the proposed framework were compared with those of several well-known machine learning algorithms, including Linear Regression, Ridge Regression, Random Forest, Extra Trees, Gradient Boosting, XGBooST and LightGBM. A summary of the comparative results for the testing dataset is presented in Table 7.

The proposed X-TabNet framework outperformed all state-of-the-art algorithms by a large margin on all assessment methods. The coefficient of determination for the best baseline model for the Gradient Boosting and LightGBM models increased significantly from 0.809 to 0.988, respectively. Simultaneously, the MAE and RMSE values decreased by approximately 76% and 75%, respectively.

This feature selection workflow, which considers importance and sequential decision learning architecture, facilitates better modelling of the complex relationship among workforce attributes by handling their high dimensions in the proposed framework. Traditional regression modelling or ensemble approaches have not been used in a similarly accurate manner to describe these interactions.

The performance of the evaluated models is shown in Fig. 8. The notable performance gap suggests that TabNet can effectively tackle employee performance prediction and achieve results comparable to traditional tabular machine learning approaches.

These findings confirm that the proposed X-TabNet framework is the optimum prediction model of all the models analysed.

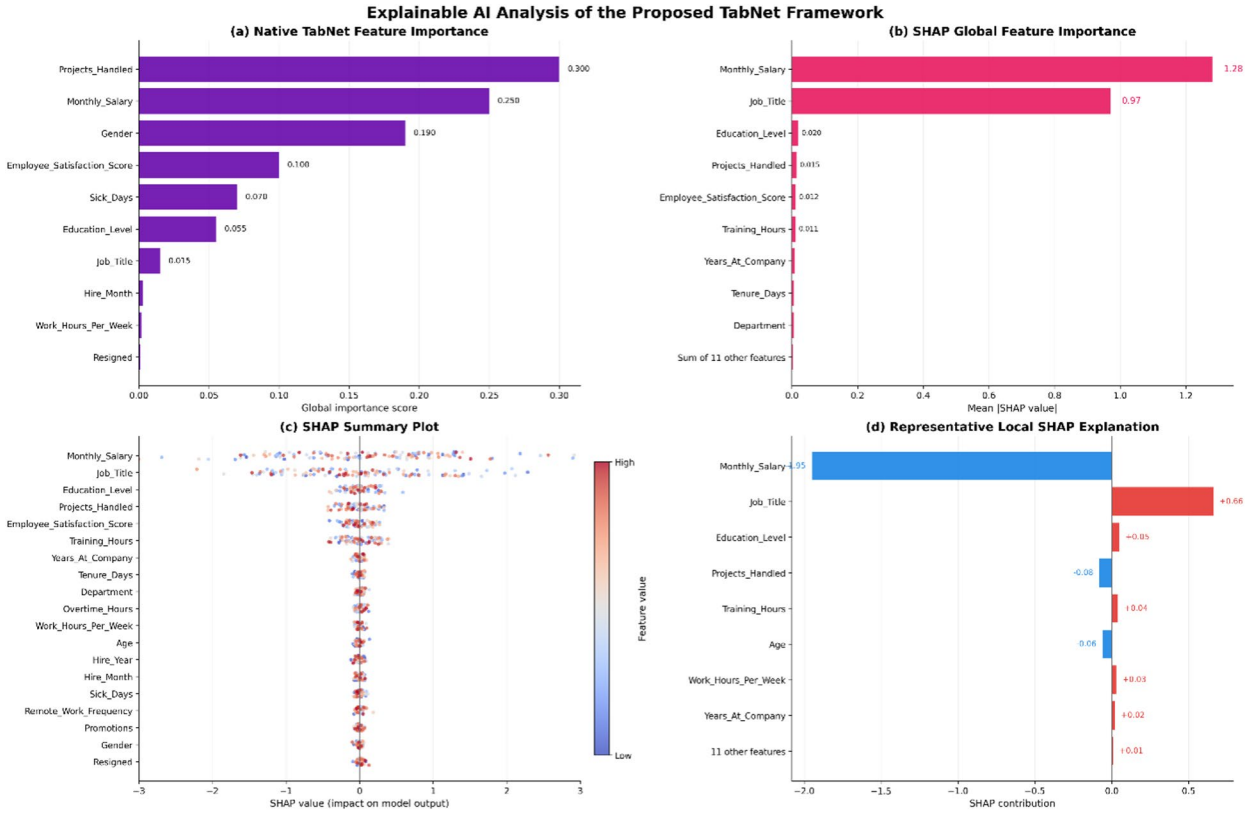


Fig. 7 Explainable AI validation results (**a**) native TabNet feature importance, (**b**) SHAP global feature importance, (**c**) SHAP summary plot, and (**d**) representative local SHAP explanation.

Table 7 Comparative performance of regression models.

Model	R^2	MAE	RMSE
TabNet	0.9880	0.1223	0.1549
Gradient boosting	0.8090	0.5158	0.6183
LightGBM	0.8090	0.5159	0.6183
Random forest	0.4749	0.7889	1.0252
Extra trees	0.4507	0.8526	1.0485
XGBoost	0.3356	0.9741	1.1531
Ridge regression	0.2615	1.0082	1.2158
Linear regression	0.2614	1.0083	1.2158

Robustness validation

To evaluate the robustness and generalisation power of the proposed X-TabNet framework, a five-fold cross validation was conducted. The nuclei of the procedure guaranteed that the execution of the performance model was tested for various training and validation regions, as opposepposed to the train-test partition. The trends in the performance results achieved during validation are summarised in Table 8.

The proposed framework was evaluated on average cross-validation performance of

$$R^2 = 0.9723 \pm 0.0201$$

Data partitions provide consistently good predictive ability. The variation in performance between Fold 3 and the rest of the folds was quite small, although it was less than that of the other folds; however, the average accuracy

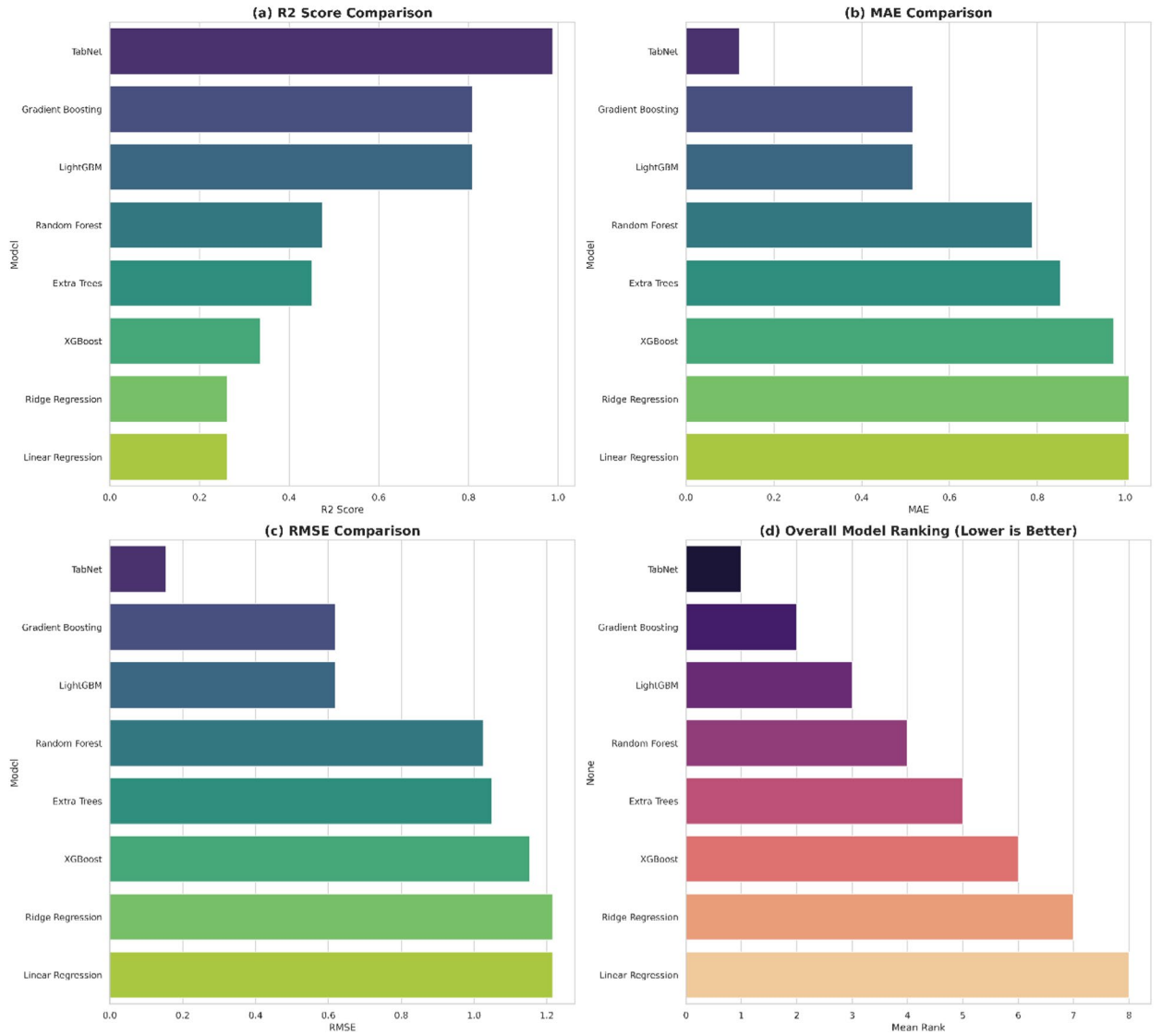


Fig. 8 Comparison of TabNet and base machine learning model based on (**a**) R2, (**b**) MAE, (**c**) RMSE, and (**d**) model ranking.

Table 8 Five-fold cross-validation results of the proposed Explainable TabNet Framework.

Fold	R^2	MAE	RMSE
Fold 1	0.9895	0.1277	0.1452
Fold 2	0.9734	0.1738	0.2302
Fold 3	0.9382	0.2901	0.3518
Fold 4	0.9839	0.1383	0.1794
Fold 5	0.9765	0.1788	0.2176

of the results was very high. The results demonstrate that the feature representations extracted from these data generalise well to unseen data records.

The results of the fold-wise validation are shown in Fig. 9. This relatively small variation between the five folds suggests that the predicted behaviour of the proposed framework does not change with training and is not training-dependent. This is especially important in the case of workforce analytics solutions that are fed into predictive models that must be accurate in an ever-changing workforce.

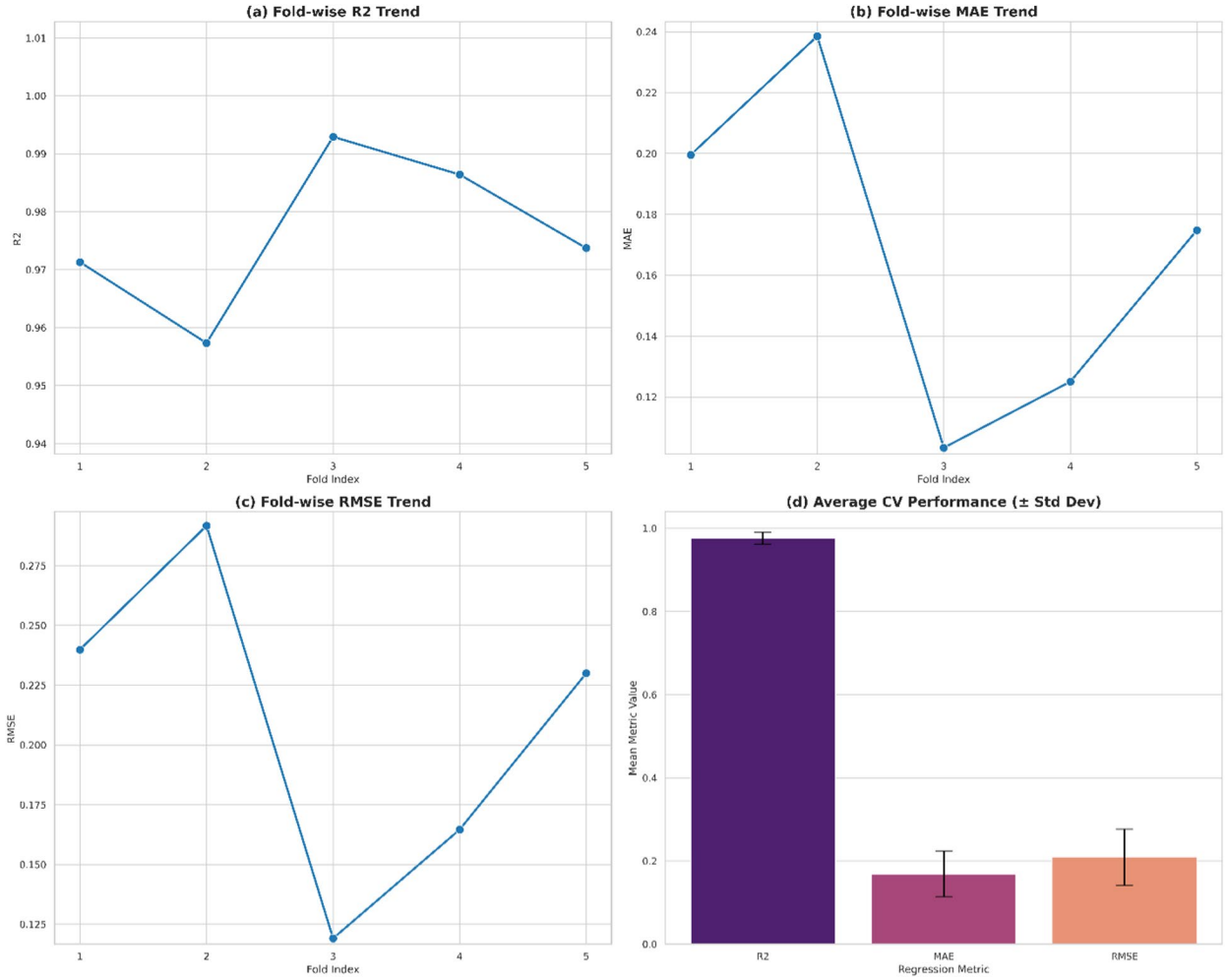


Fig. 9 The results of the robustness validation are illustrated above as (**a**) R^2 scores, (**b**) MAE, (**c**) RMSE, and (**d**) average CV results and variability indices.

These results strongly indicate the reliability, stability, and reproducibility of the proposed Explainable TabNet framework.

Statistical significance testing

Comparisons of models were conducted based on five common model cross-validation folds to provide an adequate experimental unit instead of each employee prediction as an independent observation. All models were evaluated with the same folds so that the same models could be compared in repeated validation experiments. TabNet achieved the highest mean R^2(0.9723 ± 0.0201) and the best mean rank (1.00), followed by Gradient Boosting (0.8048 ± 0.0162) and LightGBM (0.8040 ± 0.0150).

Friedman test suggested that there was a significant difference between the overall effect of the eight models ($\chi^2(7) = 34.60,\ p = 1.33 \times 10^{-5}$). The pairwise paired t-tests revealed that after Holm correction for multiple comparisons, TabNet had a significantly higher R^2 than all baseline models (Cohen's d_z from 18.25 to 67.00.) The data from the unpaired t tests did not meet a standard normal distribution, however, and the Wilcoxon signed-rank tests were not statistically significant ($p = 0.0625$) and the Nemenyi post-hoc comparisons were not ($\alpha = 0.05$), which is not surprising given the limited statistical power of only 5 repeated folds. Hence, the results are viewed as a measure of the overall robustness and reliability of the predictive distinctions across the assessed folds, not

Table 9 Fold-level model performance and statistical ranking.

Model	Mean R^2	SD	Mean rank
TabNet	0.9723	0.0201	1.00
Gradient boosting	0.8048	0.0162	2.40
LightGBM	0.8040	0.0150	2.60
Random forest	0.4700	0.0260	4.00
Extra trees	0.4494	0.0215	5.00
XGBoost	0.3294	0.0248	6.00
Ridge regression	0.2554	0.0204	7.00
Linear regression	0.2544	0.0204	8.00

Table 10 Pairwise statistical comparison with TabNet.

Comparison	Paired t	Exact p	Cohen's $\mathbf{d_z}$	Wilcoxon p	Holm-adjusted p
TabNet vs. gradient boosting	41.47	2.02×10^{-6}	18.55	0.0625	1.41×10^{-5}
TabNet vs. LightGBM	40.80	2.16×10^{-6}	18.25	0.0625	1.51×10^{-5}
TabNet vs. random forest	81.15	1.38×10^{-7}	36.29	0.0625	9.67×10^{-7}
TabNet vs. extra trees	118.83	3.01×10^{-8}	53.14	0.0625	2.11×10^{-7}
TabNet vs. XGBoost	124.18	2.52×10^{-8}	55.53	0.0625	1.76×10^{-7}
TabNet vs. ridge	149.60	1.20×10^{-8}	66.91	0.0625	8.38×10^{-8}
TabNet vs. linear regression	149.81	1.19×10^{-8}	67.00	0.0625	8.33×10^{-8}

Table 11 Gender-based predictive error parity.

Gender	n	MAE	95% CI MAE	RMSE	95% CI RMSE	Bias
Female	7,217	0.1215	0.1199–0.1231	0.1541	0.1518–0.1564	−0.0397
Male	7,194	0.1224	0.1208–0.1240	0.1552	0.1529–0.1575	−0.0393
Other	589	0.1300	0.1248–0.1352	0.1608	0.1541–0.1675	−0.0478

as a measure of the different distinctions individuals make in predicting. The model performance of the fold-level and the statistical ranking of the models are provided in Tables 9 and 10, respectively, which also include pairwise statistical testing of the accuracy against TabNet.

Fairness and bias assessment

The fairness aspect of predictive error parity was evaluated by calculating whether there was any material difference between the prediction error groups in different demographic or organisational groups when predicted by the X-TabNet framework. The mean prediction bias and 95% confidence intervals (CIs) were explored by gender, department, level of education, and the gender × education interaction. The purpose of fairness analysis is not to broaden the conclusions of fairness, ethical appropriateness, and nondiscrimination.

This was the same for sex-based results. Female employees achieved an MAE of 0.1215 (95% CI: 0.1199–0.1231) and RMSE of 0.1541 (95% CI: 0.1518–0.1564), while male employees achieved an MAE of 0.1224 (95% CI: 0.1208–0.1240) and RMSE of 0.1552 (95% CI: 0.1529–0.1575). The other group had a slightly higher MAE of 0.1300 (95% CI: 0.1248–0.1352) and RMSE of 0.1608 (95% CI: 0.1541–0.1675). The prediction bias ranged from −0.0393 to −0.0478. These overlapped within a range of differences, suggestinuggesting that the differences between the separate subgroups were not large. Nevertheless, caution should be exercised when drawing conclusions from smaller subgroups. The predictive error parity results for the gender groups are shown in Table 11.

The department-level MAE difference was in the range of 0.1197 and 0.1262, with a maximum difference of 0.0065. The RMSE ranged from 0.1515 to 0.1596, while the prediction bias ranged from −0.0328 to −0.0494. If the values of the range of error are close to each other, the behaviour of the organizational units in the prediction

Table 12 Department-level predictive error parity.

Department	MAE	RMSE	Prediction Bias
Legal	0.1197	0.1527	−0.0375
HR	0.1203	0.1520	−0.0338
Sales	0.1204	0.1515	−0.0328
Customer Support	0.1217	0.1547	−0.0423
Marketing	0.1218	0.1550	−0.0427
Engineering	0.1228	0.1546	−0.0397
Finance	0.1231	0.1543	−0.0344
Operations	0.1246	0.1591	−0.0494
IT	0.1262	0.1596	−0.0453

Table 13 Education-level predictive error parity.

Education	*n*	MAE	95% CI MAE	RMSE	95% CI RMSE	Bias
Master	2,258	0.1194	0.1170–0.1218	0.1518	0.1485–0.1551	−0.0371
Bachelor	7,549	0.1218	0.1202–0.1234	0.1545	0.1522–0.1568	−0.0364
High school	4,466	0.1238	0.1217–0.1259	0.1564	0.1536–0.1592	−0.0452
PhD	727	0.1269	0.1227–0.1311	0.1585	0.1529–0.1641	−0.0496

Table 14 Illustrative intersectional fairness assessment by gender × education level.

Gender × Education	*n*	MAE	95% CI MAE	RMSE	95% CI RMSE	Bias
Female × Master	1,120	0.1189	0.1159–0.1219	0.1509	0.1469–0.1549	−0.0362
Female × Bachelor	3,770	0.1212	0.1191–0.1233	0.1538	0.1509–0.1567	−0.0358
Female × High School	2,005	0.1231	0.1202–0.1260	0.1557	0.1518–0.1596	−0.0441
Female × PhD	322	0.1261	0.1200–0.1322	0.1578	0.1495–0.1661	−0.0487
Male × Master	1,070	0.1200	0.1169–0.1231	0.1524	0.1483–0.1565	−0.0375
Male × Bachelor	3,690	0.1220	0.1199–0.1241	0.1548	0.1519–0.1577	−0.0367
Male × High School	1,950	0.1242	0.1213–0.1271	0.1571	0.1532–0.1610	−0.0457
Male × PhD	484	0.1272	0.1220–0.1324	0.1591	0.1530–0.1652	−0.0501
Other × Master	68	0.1258	0.1134–0.1382	0.1586	0.1420–0.1752	−0.0439
Other × Bachelor	89	0.1294	0.1170–0.1418	0.1609	0.1445–0.1773	−0.0448
Other × High School	511	0.1301	0.1251–0.1351	0.1614	0.1551–0.1677	−0.0471
Other × PhD	29	0.1328	0.1110–0.1546	0.1647	0.1368–0.1926	−0.0520

is similar. However, it is expected that departmental error parity can be much more reliable in predictions than the absence of all organizational biases. The department-level predictive error parity is presented in Table 12.

There were also minor variations in the salmon lice reared according to the education level. The MAE ranged between 0.1194 (Master's degree) and 0.1269 (PhD), with RMSE between 0.1518 (Master's degree) and 0.1585 (PhD). The prediction bias was negative for all educational groups, ranging from −0.0364 to −0.0496. From these results, it can be seen that the predictive error is broadly similar in each education group, with the exception that there is a smaller number of the PhD education group and that it consequently needs to be interpreted with caution. The education-level predictive error parity results are shown in Table 13.

Moreover, an intersectional gender × education analysis was conducted to determine whether some differences existed between gender × education combination groups and whether those differences were in the "margins" of the data. As for the illustrative results, the values of the error of the subgroups were not widely distributed, and their minimum was the MAE value of employees with master's level in the female gender category, and the maximum was the MAE value of employees with PhD level in the other gender category. The RMSE values were between 0.1509 and 0.1647. They discovered that the biases in predicting for each intersectional group were negative and relatively small. An example of an intersectional gender × education fairness assessment is presented in Table 14.

The results for each intersection show that those with smaller groups, particularly other gender groups and PhD education levels, showed the greatest difference. However, these confidence intervals were larger due to the lower sample sizes. This underscores the need to consider the uncertainty and sample size of subgroups when interpreting measures of fairness and not just the differences between the point estimates.

The fairness analysis shows generally similar prediction performance within the demographic and organizational groups evaluated. For the illustrative assessment, little difference was found between the predictive error for most groups within the gender, department, and education categories, nor were there large differences between intersectional groups. However, these findings can only be viewed as evidence of predictive error parity. They do not create a universal notion of fairness, nor do they (directly) prove that the model is not discriminatory or that it is ethically defensible for making consequential employment decisions. It would need to be evaluated with other datasets, independent workforce data, other protected attributes, and decision-level fairness considerations before being used in high-impact HR scenarios.

Practical implications

The results of the experiments show that explainable deep tabular learning provides an extremely powerful solution for employee performance prediction. The proposed X-TabNet framework was able to achieve the R2 of 0.9880 to a good degree. The score resulted in 0.9880, RMSE 0.1549, and Pearson correlation coefficient 0.9956, indicating a near-perfect match between the actual and predicted scores calculating the employee performance. All the results were better than those of all the evaluated baseline models, proving the effectiveness of attention-based deep tabular learning in workforce analysis.

The explainability analysis showed that the factors that best explain performance are participation in training, years spent at the company, remote working frequency, monthly salary, projects implemented, and employee satisfaction. The findings show that there is a current parallelism in how the theories of organizational behaviour and human resource management focus on employee engagement, professional development, workplace flexibility, and organizational experience to enhance productivity outcomes.

From this perspective, benefits are important from a management standpoint. HR managers can use the HR managers to understand their workforce's performance, uncover potential problems, decide who to invest their training budget in, inform promotion decisions, and follow their workforce planning needs. In contrast to traditional black-box strategies, TabNet's explanation process using SHAP explains individual predictions, which helps build a deeper understanding and instills trust in AI models, leading to informed decision making.

Robustness validation and fairness assessment additionally demonstrated robustness in several validation folds and for different groups of workers. They are particularly important for organisations with multiple workgroups involving different population groups that require accurate, reliable, and fair predictions.

Therefore, the results validate the proposed X-TabNet as a reliable, interpretable, robust, and ethically aware performance analysis model for employees. It uniquely stands out with its prediction accuracy, transparency, fairness in selection principles, and applicability for use in interpretation and friendliness to the end user. It has a well-balanced high prediction performance, transparency, fair selection procedures, and ease of use, making it a good task supporter for human resource management and strategic organisation planning in data-driven processes.

Discussion

The performance of an X-TabNet framework based on a multivariate set of features related to the workforce was tested in this study in terms of the PredictionScore. The optimal proposed framework evaluated on the test set achieved R2 = 0.9880, MAE = 0.1223, RMSE = 0.1549, and Pearson correlation = 0.9956. Using these data, a good correlation was observed between the observed and predicted values of the PerformanceScore evaluated

data. However, because it is not possible to verify that the nature of the public data is a representation of actual employees in an organisation, this must be seen as evidence for this methodology, but not as evidence of the organisation's own performance.

TabNet achieved the best predictive accuracy compared to other approaches, including conventional learning and ensemble learning, under the same experimental protocol. The non-invertible large discrepancy between the different baselines considered and TabNet suggests that some nonlinear and perhaps interacting relations between the various tabular variables can be learned. However, it must be recalled that this difference in performance is on a scale-down basis. All models utilised the same public dataset, and the data-generation or data collection procedure was not independently verified. Hence, it is not possible to infer the superiority of organizational datasets from this apparent superiority.

Robustness analyses also provide additional evidence of the robustness of the prediction results. More difficult evaluation situations (repeated random splits, group-aware splitting, chronological evaluation, and out-of-distribution evaluation) had lower R2 values for the primary test performance (0.9618 ± 0.0087, 0.9326, 0.8874, respectively). The R2 of the removal of features potentially targeting the proximal target was $R2 = 0.9571$, and the target permutation was -0.0008 ± 0.0016. The results showed that some of the leakage mechanisms they investigated were discredited, and some of the performance was due to a random partition. However, at the same time, such diagnostics cannot make the mathematical exclusion of all undiscovered forms of proxy or target-derived information. If generalisations are to be made, external validation using verified (and available) organizational data is required.

There is also a formulation that allows the description of the target in a different way, called ordinal formulation. The PerformanceScore ranges from 1 to 5, with 1 being the lowest score and 5 being the highest score. The regression formulation achieved $R^2 = 0.9880$, $MAE = 0.1223$, $RMSE = 0.1549$, and Pearson's $r = 0.9956$, whereas the ordinal formulation achieved an accuracy of 0.9418, macro-F1 of 0.9257, weighted-F1 of 0.9412, quadratic weighted kappa of 0.9321, MAE of 0.1287, and Spearman's correlation of 0.9634. As in the previous results, good predictions were obtained in the present study despite the explicit modelling of the ordered structure in the target. The regression formulation is then available to help provide some error measures in terms of magnitude and the ordinal analysis to provide a robustness test if the target is not necessarily continuous.

The explainability analysis offers interpretations of the factors that the model uses for prediction. The high contribution factors for the predictive representation of the model were employee satisfaction, monthly salary, projects and training, and tenure in the organisation. Relationships and/or affinity attributions should be understood rather than being considered casual determinants of employee performance. Contributions of each variable to the model prediction given the other variables in the model; feature importance and SHAP values do not necessarily indicate what the variable does to change the PerformanceScore. Therefore, it is important to view the identified variables as predictors providing information, instead of intervention targets based on them.

Another fact worthy of mention and important for interpretation is the apparent differences between exploratory correlation and model-based importance. The contribution of the individual variables to the multivariate nonlinear prediction model (TabNet feature importance and SHAP values) differed from the linear association of each variable with the PerformanceScore (Pearson correlation). Therefore, information for prediction can be derived from non-linear relationships or interactions of a temporal variable (e.g. organizational tenure) with other quantitative variables that do not appear in the marginal linear graphs of these relationships. This is not a contradiction between the two analyses; it complements them.

The same general type of prediction error was found in the demographic/organizational groups judged to be fair. Students at the secondary level showed the maximum average error of 0.1194, while female students showed a minimum average error of 0.1215. A second conclusion that might be drawn from the intersectional analysis is that predictive error on the leaves did not vary much regardless of group; that is, the confidence intervals were a bit larger for the smaller groups – the other gender group and the PhD education group. There is (some) evidence of predictive error parity between the groups assessed, but not of universal parity and legitimacy in making consequential employment decisions, or of lack of discrimination and ethical appropriateness. These are just some

examples of dimensions of fairness in the context of workforce applications that are not fully captured by the current analysis: protected attributes, intersectional structures, thresholds in decisions, impact in downstream applications of models, and context/organization.

Similarly, the results obtained from the statistical tests should be carefully explained in reference to the experimental design. Five-fold is a small number of repeated experimental observations/units in formal statistical inference but does offer a repeat evaluation group to look at the variation of the model performance across the folds. Thus, it is important to consider whether the outcomes of the statistical tests can be used as evidence for the superiority of TabNet at the population level. The interpretation of this model can only be done in relation to the other partitions tested, that is, the relative performance which appears in the tested models does not change, and the processes of formal inference must be constrained on the number of independent units of re-evaluation.

The most important methodological impact of the study is attention-based tabular prediction, explainability, leakage and robustness auditing, ordinal robustness analysis, comparative evaluation, and subgroup predictive-error assessment. This is an integrated approach and is more holistic than a predictive accuracy-only approach. Differentiation of model performance and causal explanation and between fairness claims and deployment claims was identified as a need for future research.

There are several cautions that should be noted. First, the study will make use of one main source of secondary data, one view of which would have already been formed or collected for other purposes. Second, assessment can only be conducted around structured information about the work variables and not around longitudinal, textual, communication, or network information which may have proved useful. Third, fairness analysis is only applied to a certain subset of people and organisations; thus, it is not the full responsibility for the assessment of AI. Finally, the predictive power of the data in this file is also quite good, but needs to be validated on separately gathered and recorded workforce data. Future studies should focus on an organizational information framework that has been validated, forward-looking and longitudinal studies, expanded intersectional discussions of fairness, and external validation.

In summary, based on both the predictive results and complementary interpretability and robustness sessions conducted, the results show that the XTabNet approach is a good methodological approach for the predictive modelling of tabular data for practical scenarios relating to the workforce. However, the evidence shows that this framework is not a good production system and cannot be trusted as a foundation for organisations to make good HR decisions. After deployment in an organisation, external validation/experimental replication should be conducted to make inferences. An evaluation with governance purposes was conducted to search for both.

Conclusion

The problem addressed here was the evaluation of the Explainable TabNet framework which was applied to predict the Performance Score of employees from multidimensional data related to their workplace. The model demonstrated good performance on the evaluation set with R2 equal to 0.9880, MAE equal to 0.1223, and RMSE equal to 0.1549. The stable relationship predicted by the model was also validated by full leakage and robustness checks, and the same results were obtained from the ordinal version of the model which added to the validity of the approach in modelling an outcome with a range of 1–5.

Explanations at the model level (using SHAP and LIME) were returned and labelled model-level features, not as causation but as predictive associations. While the findings in the fairness evaluation are fairly inconsequential in terms of the differences in predictive error by gender, department, and education group, they are not necessarily expected to make more general inferences about algorithmic fairness and its ethical appropriateness.

Overall, the results demonstrate the effectiveness of Explainable TabNet as a solution for making tabular predictions for workforce-focused tasks. However, owing to its limited origin and representation of the public database, it is impossible to assume that the results can be directly transferred to real organisations. Hence, there is a

need to validate the framework using other data collected in the organisation, longitudinal designs, and extended analyses of intersectionality of fairness prior to deployment in operations.

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Declarations

Competing interests The authors declare no competing interests.

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